# Centering the error makes the posterior mean optimal

**Average; $m - a$ fixed** Previous slide: pointwise squared-error expansion

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

$$E[r|y] = E[z|y] - m = 0$$

$$E[r^T(m - a)|y] = E[r|y]^T(m - a) = 0$$

$$V = \mathrm{Cov}[z|y]$$

$$E[\|r\|^2|y] = \sum_{j=1}^{d} E[r_j^2|y] = \mathrm{tr}\, V$$

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**Conditional squared-error risk**

$$R(a|y) = \mathrm{tr}\, V + \|m - a\|^2$$

- Risk is residual uncertainty plus squared distance from the posterior mean.

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**Conditional squared-error risk**

$$R(a|y) = \mathrm{tr}\, V + \|m - a\|^2$$

$$\|m - a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$

$$a^*(y) = m = E[z|y]$$

$$\min_a R(a|y) = \mathrm{tr}\, V$$

Earlier Gaussian: $a^*(y) = \mu_{z|y}$

$$\min_a R(a|y) = \mathrm{tr}\, \Sigma_{z|y}$$

- Risk is residual uncertainty plus squared distance from the posterior mean.
- The posterior mean minimizes model expected squared error; covariance measures what remains.